

Lab 2: AST Printer

1. Grammar:

```
expression  → logic_or
logic_or    → logic_and ( "||" logic_and ) *
logic_and   → equality ( "&&" equality ) *
equality     → comparison ( ("!=" | "==") comparison ) *
comparison  → term ( ">" | ">=" | "<" | "<=" ) term ) *
term         → factor ( "-" | "+" ) factor ) *
factor       → unary ( "/" | "*" | "%" ) unary ) *
unary        → ("!" | "-") unary | primary
primary      → NUMBER | STRING | "true" | "false" | "nil"
              | "(" expression ")"
```

Since my language uses “&&” and “||”, I have included those in the grammar above. In addition, I have included “%”.

2. Printer Testing:

From the project root, you can run the following to compile the printer and the tests:

```
1 g++ -std=c++17 -Wall -Wextra src/token.cpp \
2 src/ast-printer.cpp test/lab2/ast-test.cpp \
3 -o ast-test
```

Then you can run the executable:

```
1 ./ast-test
```

3. **Test Cases:** The test cases cover all literal values (numbers, strings, booleans, nil), all the unary operators, binary operators, comparisons, “&&” and “||”, and larger arithmetic and logic expressions that use multiple different operators.

Here is the output I obtained from the test cases:

```
1 ./ast_test
2 PASS literal number
3   expected: 42
4   actual:   42
5 PASS literal string
6   expected: hi
7   actual:   hi
8 PASS literal true
9   expected: true
10  actual:   true
11 PASS literal false
12  expected: false
13  actual:   false
14 PASS literal nil
15  expected: nil
16  actual:   nil
17 PASS unary minus
18  expected: (- 3)
19  actual:   (- 3)
20 PASS unary bang
21  expected: (! true)
22  actual:   (! true)
23 PASS nested unary
24  expected: (! (! false))
25  actual:   (! (! false))
26 PASS arithmetic
27  expected: (- (+ 1 (* 2 3)) (% (/ 4 5) 6))
28  actual:   (- (+ 1 (* 2 3)) (% (/ 4 5) 6))
29 PASS assignment example
30  expected: (* (- 123) (group 45.67))
31  actual:   (* (- 123) (group 45.67))
32 PASS logic
33  expected: (|| (&& (== (< 1 2) true) (!= (>= 3 4) nil)) (group (&&
   ↪ (<= 5 6) (> 7 8))))
```

```
34  actual:    (|| (&& (== (< 1 2) true) (!= (>= 3 4) nil)) (group (&&
    ↪  (<= 5 6) (> 7 8))))
35
36  All tests passed.
```
